Assignment # 3 – Solution

1. (a) (i) Solution:
Suppose $a|b$ and $a|c$. Then there are integers m and n such that $b = am$ and $c = an.$
Then $b + c = am + an = a(m+n)$, so $a|(b + c)$.
(ii) Solution:
Suppose $a|b$. Then there is an integer n such than $b = an.$
Then $bc = (an)c = a(nc)$, so $a|(bc)$.
(iii) Solution:
If $a|b$ and $b|c$. Then there are integers m and n such that $b = am$ and $c = bn.$
Then $c = bn = (am)n = a(mn)$, so $a|c$.
(b) Solution:
In any two consecutive integers.
one of them will be even (2m) and the other will be odd (2m + 1)
$= 2m \times (2m + 1) = 2(2m^2 + m)$. Hence the product is even.
(c) Solution:
Given two odd integers 2m + 1 and 2n + 1 and an even integer is 2k.
= (2k)*(2m + 1)*(2n + 1) = 2(k*(2m + 1)*(2n + 1)). Hence the product is an even integer.

2. (a) Solution:
Let $\sqrt{(a+b)}$ = $\sqrt{a}+\sqrt{b}$
Squaring, we get a + b = a + b + 2 $\sqrt{a}\sqrt{b}$
⇒ $0 = 2\sqrt{a}\sqrt{b}$ cancelling a + b
⇒ $0 = 2\sqrt{ab}$
⇒ 0 = ab squaring
⇒ either a = 0 or b = 0
It means that if we want to find out the integers which satisfy the given condition then one of them must be zero.
Hence if we let a = 0 and b = 3 then
R.H.S = $\sqrt{(a+b)}$ = $\sqrt{0+3} = \sqrt{3}$
Now,
L.H.S = $\sqrt{0}+\sqrt{3} = \sqrt{3}$
From above it quite clear that the given condition is satisfied if we take a=0 and b=3.
(b) Solution:
1. Assume r is a rational number. By the definition of rational numbers, r can be written as $\frac{a}{b}$, where a and b are integers and $b \neq 0$.
$$r = \frac{a}{b}$$
2. Now, consider the square of r:
$$r^2 = \left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2}$$
3. Since the square of an integer is still an integer, a^2 and b^2 are integers. Additionally, $b^2 \neq 0$ because $b \neq 0$.
4. Therefore, $r^2 = \frac{a^2}{b^2}$ is a ratio of two integers with a non-zero denominator, which means r^2 is a rational number.

(c) Solution:
Here we are asked to show a single integer for which $2^n - 1$ is prime. First of all, we will check the integers from 1 and check whether the answer is prime or not by putting these values in $2^n - 1$. When we got the answer is prime then we will stop our process of checking the integers and we note that,
Let n = 7, then
$2^n - 1 = 27 - 1 = 128 - 1 = 127$ and we know that 127 is prime.

3. (a) Solution:
Suppose n and m are odd and n + m is not even (odd i.e., by taking contradiction).
Now n = 2p + 1 for some integer p and m = 2q + 1 for some integer q
Hence n + m = (2p + 1) + (2q + 1) = 2p + 2q + 2 = 2· (p + q + 1)
which is even, contradicting the assumption that n + m is odd.

(b) Solution:
Suppose there exists an integer a and a prime number p such that p|a and p|(a+1).
Then by definition of divisibility there exist integer r and s so that
$a = p \cdot r$ and $a + 1 = p \cdot s$
It follows that
$1 = (a + 1) - a$
$= p \cdot s - p \cdot r$
$= p \cdot (s - r)$ where $s - r \in Z$
This implies $p \mid 1$.
But the only integer divisors of 1 are 1 and -1 and since p is prime p>1. This is a contradiction. Hence the supposition is false, and the given statement is true.
(c) Solution:

Suppose the set of prime numbers is finite.
Then, all the prime numbers can be listed, say, in ascending order:
$p_1 = 2,\ p_2 = 3,\ p_3 = 5,\ p_4 = 7,\ \ldots, p_n$
Consider the integer
$N = p_1 \cdot p_2 \cdot p_3 \cdot \cdots \cdot p_n + 1$

Then $N > 1$. Since any integer greater than 1 is divisible by some prime number p, therefore $p \mid N$.
Also since p is prime, p must equal one of the prime numbers
$p_1, p_2, p_3, \cdots, p_n$.
Thus
$P \mid (p_1, p_2, p_3, \cdots, p_n)$

But then
$P \nmid (p_1, p_2, p_3, \cdots, p_n + 1)$

So $P \nmid N$

Thus $p \mid N$ and $p \nmid N$, which is a contradiction.

Hence the supposition is false and the theorem is true.

4. (a) Solution:
The contrapositive statement is:
if $x \le 1$ and $x \ge -1$ then $|x| \le 1$ for x € R.
Suppose that $x \le 1$ and $x \ge -1$
$\Rightarrow x \le 1$ and $x \ge -1$ $\Rightarrow -1 \le x \le 1$
and so $|x| \le 1$ Equivalently $|x| > 1$.
(b) Solution:
"For all integers m and n, if m and n are not both even and m and n are not both odd, then m + n is not even."
Or more simply, "For all integers m and n, if one of m and n is even and the other is odd, then m + n is odd"
Suppose m is even and n is odd. Then
$m = 2p$ for some integer p and $n = 2q + 1$ for some integer q
Now $m + n = (2p) + (2q + 1) = 2 \cdot (p + q) + 1 = 2 \cdot r + 1$ where $r = p + q$ is an integer. Hence m + n is odd.
Similarly, taking m as odd and n even, we again arrive at the result that m + n is odd. Thus, the contrapositive statement is true. Since an implication is logically equivalent to its contrapositive so the given implication is true.
(c) Solution:
Contraposition: if a and b are positive, even integers, then ab is divisible by 4.
Suppose that a = 2k and b = 2k. Then, we have
$Ab = (2k)(2k) = 4k^2$
Hence, ab is divisible by 4.

5. (a) (i) Solution: Let the prime number n be 7, then $n + 2 = 7 + 2 = 9$ which is not prime.
(ii) Solution: Consider the even number 6.
$6 \div 4 = 1.5$ Since 6 is not divisible by 4, this disproves the statement.

(b) (i) Solution:

Consider the positive integer $x = 4$. 4 is not a prime number since it has divisors other than 1 and itself (i.e., 2).

(ii) Solution: For all real numbers x, $x^2 > x$.

Counterexample: Consider the real number $x = 0$.

$x^2 = 0^2 = 0$ and $x = 0$. Here, $x^2 = 0$ is not greater than $x=0$. This disproves the statement.

Similarly, consider $x = 1$:

$x^2 = 1^2$ and $x = 1$ Again, $x^2 = 1$ is not greater than $x=1$.

(c) Solution:

We know that $\sqrt{2}$ is an irrational number. Now $(\sqrt{2})(\sqrt{2}) = \frac{2}{1}$ which is a rational number. Hence the statement is disproved.

6. (a) Solution:

Suppose $6-7\sqrt{2}$ is rational.

Then by definition of rational,

$$6-7\sqrt{2} = \frac{a}{b}$$

for some integers a and b with $b \neq 0$.

Now consider,

$$7\sqrt{2} = 6-\frac{a}{b}$$

$$\Rightarrow \quad 7\sqrt{2} = \frac{6b-a}{b}$$

$$\Rightarrow \quad \sqrt{2} = \frac{6b-a}{7b}$$

Since a and b are integers, so are 6b-a and 7b and $7b \neq 0$;

hence $\sqrt{2}$ is a quotient of the two integers 6b-a and 7b with $7b \neq 0$.

Accordingly, $\sqrt{2}$ is rational (by definition of rational).

This contradicts the fact because $\sqrt{2}$ is irrational.

Hence our supposition is false and so $6-7\sqrt{2}$ is irrational.

(b) Solution:

Suppose $\sqrt{2}+\sqrt{3}$ is rational. Then, by definition of rational, there exists integers a and b with $b \neq 0$ such that

$$\sqrt{2}+\sqrt{3} = \frac{a}{b}$$

Squaring both sides, we get

$$2+3+2\sqrt{2}\sqrt{3} = \frac{a^2}{b^2}$$

$$\Rightarrow \quad 2\sqrt{2\times 3} = \frac{a^2}{b^2}-5$$

$$\Rightarrow \quad 2\sqrt{6} = \frac{a^2-5b^2}{b^2}$$

$$\Rightarrow \quad \sqrt{6} = \frac{a^2-5b^2}{2b^2}$$

Since a and b are integers, so are therefore $a^2 - 5b^2$ and $2b^2$ with $2b^2 \neq 0$. Hence $\sqrt{6}$ is the quotient of two integers $a^2 - 2b^2$ and $2b^2$ with $2^2 \neq 0$. Accordingly, $\sqrt{6}$ is rational. But this is a contradiction, since $\sqrt{6}$ is not rational. Hence our supposition is false and so $\sqrt{2}+\sqrt{3}$ is irrational.

<u>REMARK:</u>

The sum of two irrational numbers need not be irrational in general for

$(6-7\sqrt{2})+(6+7\sqrt{2}) = 6+6 = 12$

which is rational.

(c) Solution:
Let z is any integer. [We must show that (4(z2 + z + 1) – 3z2) is a perfect square.] Then, we have
$4(z^2 + z + 1) - 3z^2 = 4z^2 + 4z + 4 - 3z^2 = z^2 + 4z + 4 = (z + 2)^2$
Hence, it's a perfect square because (z +2) is an integer, being a sum of z and 2.

7. (a) Solution:
<u>Basis step: n=1</u>

$n^2 = \frac{n(n+1)(2n+1)}{6}$ $1 = \frac{1(1+1)(2+1)}{6}$ $1 = \frac{6}{6} = 1$ 1 = 1 Hence P(n) is true for n=1.

<u>Inductive step</u>
n = k
$P(k) = 1+2+3 + \ldots + k^2 = \frac{k(k+1)(2k+1)}{6}$
<u>n = k+1</u>
Since $P(k) = 1+2+3 + \ldots + k^2 = \frac{k(k+1)(2k+1)}{6}$
Add $(k+1)^2$ both side

$$1+2+3+\ldots + k + (k+1) = (k+1)^2 + \frac{k(k+1)(2k+1)}{6}$$
$$= (k+1)\left(\frac{6(k+1)+k(2k+1)}{6}\right)$$
$$= (k+1)\left(\frac{6k+6+2k^2+k}{6}\right)$$
$$= (k+1)\left(\frac{2k^2+7k+6}{6}\right) = (k+1)\left(\frac{2k^2+4k+3k+6}{6}\right)$$
$$= (k+1)\left(\frac{2k(k+2)+3(k+2)}{6}\right) = (k+1)\left(\frac{(2k+3)(k+2)}{6}\right)$$
$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

Hence p(n) is true for n = k +1.

Rough work

$n^2 = \frac{n(n+1)(2n+1)}{6}$

$(k+1)^2 = \left(\frac{(k+1)(k+2)(2k+3)}{6}\right)$

(b) Solution:
$1^3 + 2^3 + 3^3 + \ldots + n^3 = ¼\, n^2(n + 1)^2$
1. Show it is true for n=1
$1^3 = ¼ \times 1^2 \times 2^2$ is True

2. Assume it is true for n=k
$1^3 + 2^3 + 3^3 + \ldots + k^3 = ¼k^2(k + 1)^2$ is True (An assumption!)
Now, prove it is true for "k+1"
$1^3 + 2^3 + 3^3 + \ldots + (k + 1)^3 = ¼\,(k + 1)^2(k + 2)^2$

3. We know that $1^3 + 2^3 + 3^3 + \ldots + k^3 = ¼k^2(k + 1)^2$ (the assumption above), so we can do a replacement for all but the last term:
$¼k^2(k + 1)^2 + (k + 1)^3 = ¼\,(k + 1)^2(k + 2)^2$
Multiply all terms by 4:
$k^2(k + 1)^2 + 4(k + 1)^3 = (k + 1)^2(k + 2)^2$
All terms have a common factor $(k + 1)^2$, so it can be canceled:
$k^2 + 4(k + 1) = (k + 2)^2$
And simplify:
$k^2 + 4k + 4 = k^2 + 4k + 4$
They are the same! So it is true.
So: $1^3 + 2^3 + 3^3 + \ldots + (k + 1)^3 = ¼\,(k + 1)^2(k + 2)^2$ is True.

(c) Solution:

SOLUTION:

Let P(n) be the given equation.

1. **Basis Step:**

   P(1) is true

   For n = 1

   L.H.S of P(1) $= \frac{1}{1\cdot 2} = \frac{1}{1\times 2} = \frac{1}{2}$

   R.H.S of P(1) $= \frac{1}{1+1} = \frac{1}{2}$

   Hence P(1) is true

2. **Inductive Step:**

   Suppose P(k) is true, for some integer k≥1. That is

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{k(k+1)}=\frac{k}{k+1} \quad \ldots\ldots(1)$$

To prove P(k+1) is true. That is

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{(k+1)(k+1+1)}=\frac{k+1}{(k+1)+1} \quad \ldots\ldots\ldots(2)$$

Now we will consider the L.H.S of the equation (2) and will try to get the R.H.S by using equation (1) and some simple computation.

Consider LHS of (2)

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{(k+1)(k+2)}$$

$$=\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)}$$

$$=\frac{k}{k+1}+\frac{1}{(k+1)(k+2)}$$

$$=\frac{k(k+2)+1}{(k+1)(k+2)}$$

$$=\frac{k^2+2k+1}{(k+1)(k+2)}$$

$$=\frac{(k+1)^2}{(k+1)(k+2)}$$

$$=\frac{k+1}{(k+2)}$$

$$= \text{RHS of (2)}$$

Hence P(k+1) is also true and so by Mathematical induction the given equation is true for all integers n ≥1.

8. (a)

(i) Solution:	q = 2;	r = 5	(ii) Solution:	q = -11;	r = 10
(iii) Solution:	q = 34;	r = 7	(iv) Solution:	q = 77;	r = 0
(v) Solution:	q = 0;	r = 10	(vi) Solution:	q = 0;	r = 3
(vii) Solution:	q = -1;	r = 2	(viii) Solution:	q = 4;	r = 0

(b)

	$q = a$ div m		$r = a$ mod m
i) a = −111, m = 99.	Solution: −2 = −111 div 99	;	87 = −111 div 99
ii) a = −9999, m = 101.	Solution: −99 = −9999 div 101	;	0 = −9999 div 101
iii) a = 10299, m = 999.	Solution: 10 = 10299 div 999	;	309 = 10299 div 999
iv) a = 123456, m = 1001.	Solution: 123= 123456 div 1001	;	333 = 123456 div 1001

(c)

i) Solution:

As We know that $a \equiv b(mod\, m)$ iff $\frac{a-b}{m}$. Now $80 \not\equiv 5\ (mod\, 17)$ because $\frac{80-5}{17} = 4.41.$

ii) Solution:

As We know that $a \equiv b(mod\, m)$ iff $\frac{a-b}{m}$. Now $103 \not\equiv 5\ (mod\, 17)$ because $\frac{103-5}{17} = 5.76.$

iii) Solution:

As We know that $a \equiv b(mod\, m)$ iff $\frac{a-b}{m}$. Now $-29 \equiv 5\ (mod\, 17)$ because $\frac{-29-5}{17} = -2.$

iv) Solution:

As We know that $a \equiv b(mod\, m)$ iff $\frac{a-b}{m}$. Now $-122 \not\equiv 5\ (mod\, 17)$ because $\frac{-122-5}{17} = -7.47.$

9. (a) (i) Solution: Yes gcd (11,15) = 1, gcd (11,15) = 1, gcd (11,19) = 1
(ii) Solution: No gcd (14, 15) = 1, gcd (14, 21) = 7, gcd (15, 21) = 3
iii) Solution: Yes. gcd (12, 17) = 1, gcd (12, 31) = 1, gcd (12, 37) = 1, gcd (17, 31) = 1, gcd (17, 37) = 1, gcd (31, 37) = 1
iv) Solution: Yes. gcd (7, 8) = 1, gcd (7, 9) = 1, gcd (7, 11) = 1, gcd (8, 9) = 1, gcd (8, 11) = 1, gcd (9, 11) = 1
(b)
i) Solution: $88 = 2^3 * 11$ ii) Solution: $126 = 2 * 3^2 * 7$ iii) Solution: $729 = 3^6$
iv) Solution: 1001 = 7 * 13 * 11 v) Solution: 1111 = 11 * 101 vi) Solution: $909 = 3^2 * 101$
(c)
(i) Solution:
Prime factorization of the numbers: 11 = 11; 15 = 3 × 5; 105 = 3 × 5 × 7
LCM (11, 15, 105) = 3 × 5 × 7 × 11 = 1155; GCD (11, 15, 105) = 1
(ii) Solution:
Prime factorization of the numbers: 14 = 2 × 7; 15 = 3 × 5; 21 = 3 × 7
LCM (14, 15, 21) = 2 × 3 × 5 × 7 = 210; GCD (14, 15, 21) = 1
(iii) Solution:
Prime factorization of the numbers: 12 = 2 × 2 × 3; 72 = 2 × 2 × 2 × 3 × 3; 31 = 31; 32 = 2 × 2 × 2 × 2 × 2
LCM (12, 72, 31, 32) = 2 × 2 × 2 × 2 × 2 × 3 × 3 × 31 = 8928; GCD (12, 72, 31, 32) = 1
(iv) Solution:
Prime factorization of the numbers: 17 = 17; 18 = 2 × 3 × 3; 192 = 2 × 2 × 2 × 2 × 2 × 2 × 3; 1021 = 1021
LCM (17, 18, 192, 1021) = 2 × 2 × 2 × 2 × 2 × 2 × 3 × 3 × 17 × 1021 = 9997632; GCD (17, 18, 192, 1021) = 1

10. (a)
(i) Solution: Gcd(144,89) = (144)(34) + (89)(−55) = 1
(ii) Solution: Gcd (1001, 100001) = (10)(100001) + (−999)(1001) = 11
(b)
(i) Solution:
Gcd(55,89) = (55)(34) + (89)(−21) = 1
So, inverse ā = 34.
Multiply 34 both side
55 * 34 x ≡ 34 * 34 (mod 89)
x = 1156 (mod 89) = 88.
(ii) Solution:
Gcd(89,232) = (73)(89) + (232)(−28) = 1
So, inverse ā = 73,
Multiply 73 both side
89 * 73 x ≡ 2 * 73 (mod 232)
x = 146 (mod 232) = 146.
(c)
(i) Solution: gcd(2,17) = (1)(17) + (−8)(2) = 1 So, −8 + 17 =9 Hence inverse, ā = 9.
(ii) Solution: gcd(34,89) = (13)(89) + (−34)(34) = 1 So, −34 + 89 = 55 Hence inverse, ā = 55.
(iii) Solution: gcd(144,233) = (89)(144) + (−55)(233) = 1 Hence inverse, ā = 89.
(iv) Solution: gcd(200,1001) = (1)(1001) + (−5)(200) = 1 So, −5 + 1001 = 996 Hence inverse, ā = 996.

11. (a)
(i) Solution:
$X_{10} \equiv 1\cdot1 + 2\cdot2 + 3\cdot5 + 4\cdot9 + 5\cdot7 + 6\cdot3 + 7\cdot 1 + 8\cdot2 + 9\cdot8 \pmod{11}$
$\equiv 1+4+15+36+35+18+7+16+72 \pmod{11} \equiv 204 \bmod 11$
$X_{10} \equiv 6$. Yes, it's an original copy since 204 +(10*6) ≡ 264 mod 11 ≡ 0.
(ii) Solution:
a = q d + r
26 = 3. 7 + 5
7 = 1.5 + 2
5 = 2.2 + 1 Now, 1 = 1.5 – 2.2; 2 = 1.7 – 1.5; 5 = 1.26 – 3.7

$1 = 1.5 + 2(1.7 - 1.5) = 1.5 - 2.7 + 2.5;$ $1 = 3.5 - 2.7 = 3(1.26 - 3.7) - 2.7$

$1 = 3.26 - 9.7 - 2.7 = 3.26 - 11.7;$ $1 = (3)(26) + (-11)(7)$

Inverse can't be negative so – 11 +26 = 15.

(b)

(i) Solution: E: $C = 04^3 \bmod 33$. X: $C = 23^3 \bmod 33$. A: $C = 00^3 \bmod 33$. M: $C = 12^3 \bmod 33$.

(ii) Solution: Encrypted Message: "STOP TALKING"

(c)

(i) Solution: -76, -51, -26, -1, 24, 49, 74, 99 are the integers.

(ii) Solution: By Fermat's little theorem, $5^{58} \equiv 1 \pmod{59}$, so $5^{116} \equiv 1 \pmod{59}$, so $5^{119} \equiv 5^3 \equiv 7 \pmod{59}$.

12. (a) (i) Solution:

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 = 5 * 6 * 7 = 210$.

$M_1 = 210/5 = 42$, $M_2 = 210/6 = 35$, and $M_3 = 210/7 = 30$

Also, by simple inspection we see that:

$y_1 = 3$ is an inverse for $M_1 = 42$ modulo 5, $y_2 = 5$ is an inverse for $M_2 = 35$ modulo 6 and

$y_3 = 4$ is an inverse for $M_3 = 30$ modulo 7.

The solutions to the system are then all numbers x such that $x = (a_1M_1y_1 + a_2M_2y_2 + a_3M_3y_3) \bmod m$

$= ((1 * 42 * 3) + (2*35*5) + (3*30*4)) \bmod 210 = 836 \pmod{210} = 206$.

(ii) Solution:

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 * m_4 = 2 * 3 * 5 * 11 = 330$.

$M_1 = 330/2 = 165$, $M_2 = 330/3 = 110$, $M_3 = 330/5 = 66$ and $M_4 = 330/11 = 30$

Also, by simple inspection we see that:

$y_1 = 1$ is an inverse for $M_1 = 165$ modulo 2, $y_2 = 2$ is an inverse for $M_2 = 110$ modulo 3,

$y_3 = 1$ is an inverse for $M_3 = 66$ modulo 5 and $y_4 = 7$ is an inverse for $M_4 = 30$ modulo 11.

The solutions to the system are then all numbers x such that $x = (a_1M_1y_1 + a_2M_2y_2 + a_3M_3y_3 + a_4M_4y_4) \bmod m$

$= ((1 * 165 * 1) + (2 * 110 * 2) + (3 * 66 * 1) + (4 * 30 * 7)) \bmod 330 = 1643 \pmod{330} = 323$.

(b) Solution:

$x \equiv 3 \pmod 5$, $x \equiv 3 \pmod 6$, $x \equiv 1 \pmod 7$, and $x \equiv 0 \pmod{11}$.

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 * m_4 = 5*6*7*11 = 2310$.

Also, by simple inspection we see that:

$y_1 = 3$ is an inverse for $M_1 = 462$ modulo 5, $y_2 = 1$ is an inverse for $M_2 = 385$ modulo 6,

$y_3 = 1$ is an inverse for $M_3 = 330$ modulo 7 and $Y_4 = 1$ is an inverse for $M_3 = 210$ modulo 11.

The solutions to the system are then all numbers x such that $x = (a_1M_1y_1 + a_2M_2y_2 + a_3M_3y_3 + a_4M_4y_4) \bmod m$

$= (3 * 462 * 3) + (3 * 385 * 1) + (1 * 330 * 1) + (0*210*1) = 5643 \pmod{2310} = 1023$.

He could have 1023 oranges.

(c) Solution:

$x \equiv 3 \pmod 4$, $x \equiv 5 \pmod 9$, and $x \equiv 8 \pmod{13}$

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 = 4*9*13 = 468$.

Also, by simple inspection we see that:

$y_1 = 1$ is an inverse for $M_1 = 117$ modulo 4, $y_2 = 7$ is an inverse for $M_2 = 52$ modulo 9,

$y_3 = 4$ is an inverse for $M_3 = 36$ modulo 13.

The solutions to the system are then all numbers x such that $x = (a_1M_1y_1 + a_2M_2y_2 + a_3M_3y_3) \bmod m$

$= (3 * 117 * 1) + (5 * 52 * 4) + (8 * 36 * 4) = 2543 \pmod{468} = 203$. He has received 203 presents.

13. (a) (i) Solution: The encrypted message will be "L ORYH GLVFUHWH PDWKHPDWLFV "

(ii) (i) Solution:

S	T	O	P	P	O	L	L	U	T	I	O	N
18	19	14	15	15	14	11	11	20	19	8	14	13

After applying function:

22	23	18	19	19	18	15	15	24	23	12	18	17
W	X	S	T	T	S	P	P	Y	X	M	S	R

will be encrypted message.

ii) Solution:

S T O P P O L L U T I O N

18 19 14 15 15 14 11 11 20 19 8 14 13

After applying function:

13 14 09 10 10 09 06 06 15 14 03 09 08

N O J K K J G G P O D J I will be encrypted message.

(b) (i) Solution: “MID TWO ASSIGNMENT” will be decrypted message.

(ii)Solution: “FAST NUCES UNIVERSITY “will be decrypted message.

(c) (i) Solution: “SURRENDER NOW” will be decrypted message.

(ii) Solution: “BE MY FRIEND” will be decrypted message.

14. (a)

(i) Solution: 034567981 mod 97 = 91

(ii) Solution: 183211232 mod 97 = 57

(iii) Solution: 220195744 mod 97 = 21

(iv) Solution: 987255335 mod 97 = 5

(b)

(i) Solution: 104578690 mod 101 = 58.

(ii) Solution: 432222187 mod 101 = 60.

(iii) Solution: 372201919 mod 101 = 32.

(iv) Solution: 501338753 mod 101 = 3.

(c)

(i) Solution:

$X_1 = (4 * 3 + 1) \bmod 7 = 6.$

$X_2 = (4 * 6 + 1) \bmod 7 = 4.$

$X_3 = (4 * 4 + 1) \bmod 7 = 3.$

$X_4 = (4 * 3 + 1) \bmod 7 = 6.$

$X_5 = (4 * 6 + 1) \bmod 7 = 4$ Sequence: 6,4,3,6,4…….

(ii)Solution:

$X_1 = (7 * 5 + 2) \bmod 9 = 1.$

$X_2 = (7 * 1 + 2) \bmod 9 = 0.$

$X_3 = (7 * 0 + 2) \bmod 9 = 2.$

$X_4 = (7 * 2 + 2) \bmod 9 = 7.$

$X_5 = (7 * 7 + 2) \bmod 9 = 6$ Sequence: 1,0,2,7,6…….

15. (a) (i) Solution:

$7*3 + 3 + 2*3 + 3 + 2*3 + 1 + 8*3 + 4 + 4*3 + 3 + 4*3 + x_{12} = 0 \bmod 10$

$21 + 3 + 6 + 3 + 6 + 1 + 24 + 4 + 12 + 3 + 12 + x_{12} = 0 \bmod 10$

$95 + x_{12} = 0 \bmod 10$ Check digit is $x_{12} = 5$.

(ii) Solution:

$6*3 + 3 + 6*3 + 2 + 3*3 + 9 + 9*3 + 1 + 3*3 + 4 + 6*3 + x_{12} = 0 \bmod 10$

$18 + 3 + 18 + 2 + 9 + 9 + 27 + 1 + 9 + 4 + 18 + x_{12} = 0 \bmod 10$

$118 + x_{12} = 0 \bmod 10$ Check digit is $x_{12} = 2$.

(b) (i) Solution:

$0*3 + 3 + 6*3 + 0 + 0*3 + 0 + 2*3 + 9 + 1*3 + 4 + 5*3 + 2 = 0 \bmod 10$

$0 + 3 + 18 + 0 + 0 + 0 + 6 + 9 + 3 + 4 + 15 + 2 = 0 \bmod 10$

$60 \equiv 0 \bmod 10$ It’s a valid UPC code.

(ii) Solution:

$0*3 + 1 + 2*3 + 3 + 4*3 + 5 + 6*3 + 7 + 8*3 + 9 + 0*3 + 3 = 0 \bmod 10$

$0 + 1 + 6 + 3 + 12 + 5 + 18 + 7 + 24 + 9 + 0 + 3 = 0 \bmod 10$

$88 \not\equiv 0 \bmod 10$ It’s not a valid UPC code.

(c) (i) Solution:

$1*0 + 2*0 + 3*7 + 4*1 + 5*1 + 6*9 + 7*8 + 8*8 + 9*1 + x_{10} = 0 \bmod 11$

$0 + 0 + 21 + 4 + 5 + 54 + 56 + 64 + 9 + x_{10} = 0 \bmod 11$

$213 + x_{10} = 0 \bmod 11$ Check digit, $x_{10} = 4$.

(ii)Solution:

$$\begin{aligned} x_{10} &= 1 \cdot 0 + 2 \cdot 3 + 3 \cdot 2 + 4 \cdot 1 + 5 \cdot 5 + 6 \cdot 0 + 7 \cdot 0 + 8 \cdot Q + 9 \cdot 1 \bmod 11 \\ &= 0 + 6 + 6 + 4 + 25 + 0 + 0 + 8Q + 9 \bmod 11 \\ &= 8Q + 50 \bmod 11 \end{aligned}$$

The check digit is known to be 8.

$$8Q + 50 \bmod 11 = 8$$

Since $50 \bmod 11 = 6$

$$8Q + 6 \bmod 11 = 8$$

Subtract 6 from each side of the equation:

$$8Q \bmod 11 = 2$$

Since the inverse of $8 \bmod 11$ is $7 \bmod 11$, we should multiply both sides of the equation by 7:

$$\begin{aligned} 7 \cdot 8Q \bmod 11 &= 7 \cdot 2 \bmod 11 \\ 56Q \bmod 11 &= 14 \bmod 11 \\ Q \bmod 11 &= 3 \end{aligned}$$

Since Q is a digit (between 0 and 9), Q then has to be equal to 3.

16. (a) Solution:

A T T A C K

00 19 19 00 02 10

- **$n = 43 \cdot 59 = 2537$; $e = 13$ Encryption Function: $C = M^e \bmod n$**

C(AT)= $0019^{13} \bmod 2537$ C(TA)= $1900^{13} \bmod 2537$ C(CK)= $0210^{13} \bmod 2537$

E X A M I N A T I O N

04 23 00 12 08 13 00 19 08 14 13

- **$n = 43 \cdot 59 = 2537$; $e = 13$ Encryption Function: $C = M^e \bmod n$**

C(EX)= $0423^{13} \bmod 2537$ C(AM)= $0012^{13} \bmod 2537$ C(IN)= $0813^{13} \bmod 2537$

C(AT)= $0019^{13} \bmod 2537$ C(IO)= $0814^{13} \bmod 2537$ C(N)= $0013^{13} \bmod 2537$

(b) Solution:

A S S I G N M E N T

00 18 18 08 06 13 12 04 13 19

- **$n = 13 \cdot 19 = 247$ $e = 47$ Encryption Function: $C = M^e \bmod n$**

C(A)= $00^{47} \bmod 247$ C(S)= $18^{47} \bmod 247$ C(I)= $08^{47} \bmod 247$ C(G)= $06^{47} \bmod 247$

C(N)= $13^{47} \bmod 247$ C(M)= $12^{47} \bmod 247$ C(E)= $04^{47} \bmod 247$ C(T)= $19^{47} \bmod 247$

S E M E S T E R

18 04 12 04 18 19 04 17

- **$n = 13 \cdot 19 = 247$ $e = 47$ Encryption Function: $C = M^e \bmod n$**

C(S)= $18^{47} \bmod 247$ C(E)= $04^{47} \bmod 247$ C(M)= $12^{47} \bmod 247$ C(T)= $19^{47} \bmod 247$ C(R)= 17^{47} mod

(c) (i) Solution: Since $5^6 = 1 \bmod 7$; $(5^6)^{333} .5^5 \bmod 7 = 5^5 \bmod 7 = 3$.

(ii) Solution: Since $5^{10} = 1 \bmod 11$; $(5^{10})^{2000} .5^3 \bmod 11 = 5^3 \bmod 11 = 4$.

(iii) Solution: Since $5^{12} = 1 \bmod 13$; $(5^{12})^{166} .5^{11} \bmod 13 = 5^{11} \bmod 13 = 8$.

(iv) Solution: Since $5^{16} = 1 \bmod 17$; $(5^{16})^{126} .5^3 \bmod 17 = 5^3 \bmod 17 = 6$.

17. (a) Solution:

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 = 2261$.

Also, by simple inspection we see that:

$y_1 = 1$ is an inverse for $M_1 = 323$ modulo 7,

$y_2 = 11$ is an inverse for $M_2 = 133$ modulo 17, and

$y_3 = 4$ is an inverse for $M_3 = 119$ modulo 19.

The solutions to the system for all numbers x such that $x = (a_1M_1y_1 + a_2M_2y_2 + a_3M_3y_3) \bmod m$

$x = ((2 * 323 * 1) + (3 * 133 * 11) + (5 * 119 * 4)) \bmod 2261 = 7415 \pmod{2261} = 632$. He would have 632 balls.

(b) Solution: six hundred and thirty two

f (p) = (p + 3) mod 26

s i x h u n d r e d a n d t h i r t y t w o

18 8 23 7 20 13 3 17 4 3 0 13 3 19 7 8 17 19 24 19 22 14

Encrypted Text: VLA KXQGUHG DQG WKLUWB WZR

(c) Solution: Number = 632 so x = 6+3+2 = 11.

By Fermat's little theorem, we know that $11^{10} \equiv 1 \pmod 7$, and so $(11^{10})^k \equiv 1 \pmod 7$, for every positive integer *k*. Therefore,

$11^{302} = 11^{30\cdot 10+2} = (11^{10})^{30}.11^2 \equiv (1)^{30} \cdot 121 \equiv 2 \pmod 7$. Hence, $11^{302} \bmod 7 = 2$.

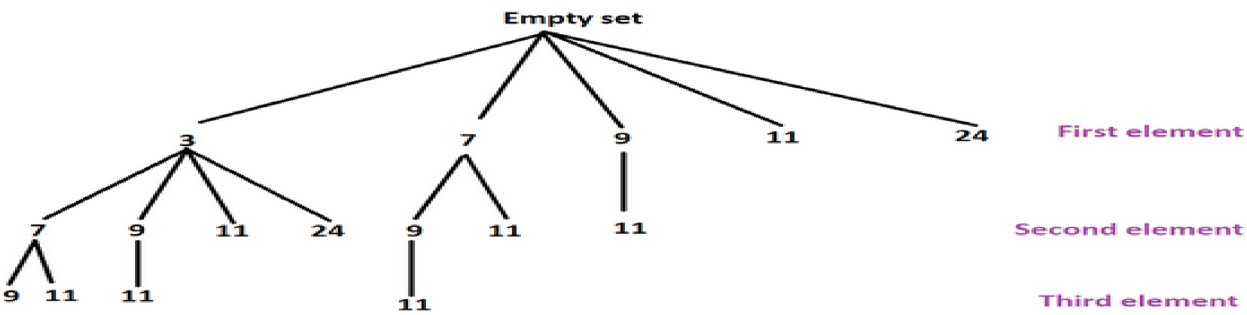
18. (a) Solution: There are 27 * 37 = 99 offices in the building.
(b) Solution: 12 * 2 * 3 shirts are required.
(c) Solution: People can have 26 *26 * 26 = 26^3 different three-letter initials.

19. (a)Solution: There are $^{13}C_5$ = 1287 ways to select the players.
(b) Solution: There are ways $^{75}C_5$ = 2071126800 to place the people on different positions.
(c) Solution: There are $^{27}C_5$ = 17,550 combinations to take the ride.

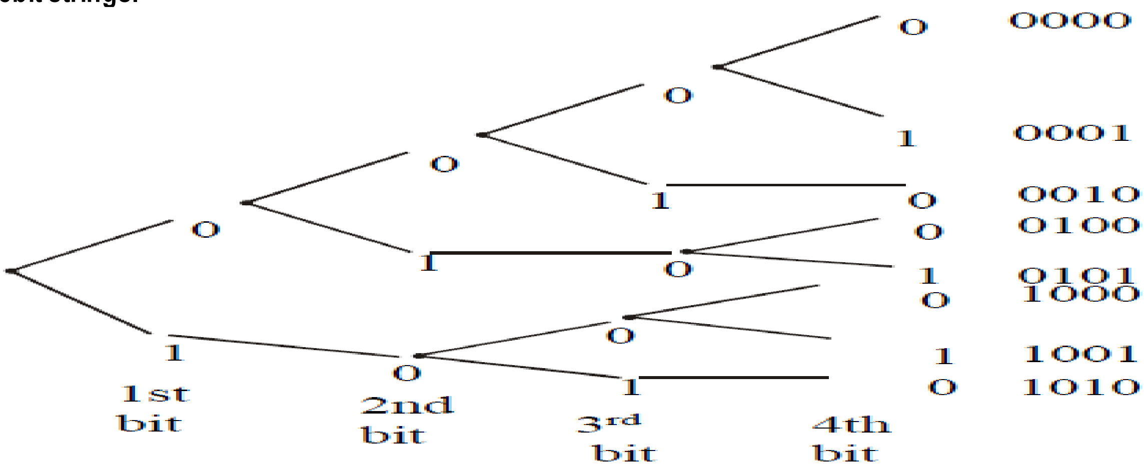
20. (a) Solution: There are 16 place values for hexadecimal numbers: 0 to 9, A, B, C, D, E and F.
So, $16^{10} + 16^{28} + 16^{58}$ different WEP keys are possible.
(b) Solution: There would be $26^4 - 25^4$ = 66,351 strings.
(c) Solution: People can have 26 *25 * 24 = 15,600 different three-letter initials with none of the letters repeated.

21. (a) Solution: Since each value of the domain can be mapped to one of two values.
Number of functions are: 2 * 2 * 2 * 2 * ... * m = 2^m.
(b) Solution: Each successive element from the domain will have one option than its predecessor as it is one-to-one function. So, number of functions are 5 * 4 * 3 * 2 * 1 = 120.
(c) Solution: There are 15 * 48 * 24 * 34 * 28 * 28 = 460,615,680 different faces.

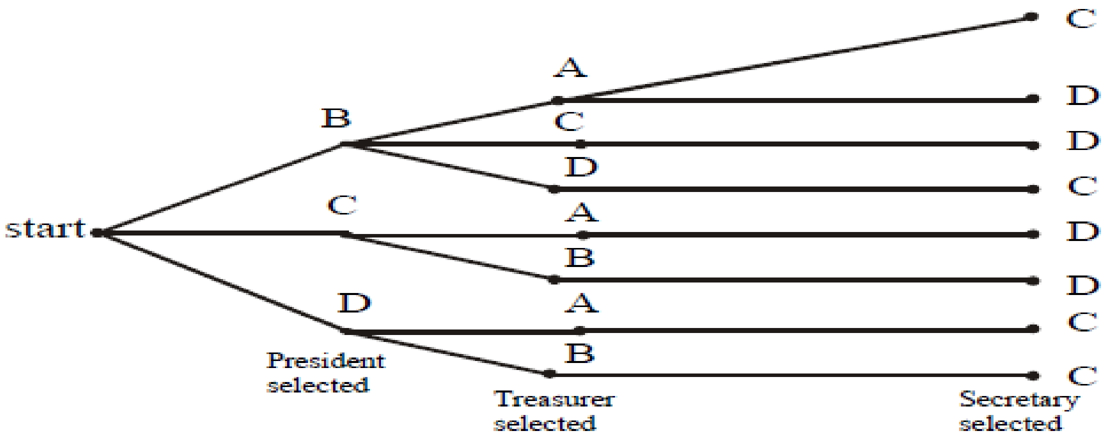
22. (a) Solution:



(b) Solution: The following tree diagrams displays all bit strings of length four without two consecutive 1's. Clearly, there are 8bit strings.



(c) Solution: We construct the possibility tree to see all the possible choices. From the tree given below, see that there are only eight ways possible to choose the offices under given conditions.



23. (a) Solution: There are ${}^{8}C_{3} = 56$ ways to choose the students.
(b) Solution: There are ${}^{12}C_{6} = 924$ ways to select the elective courses.
(c) Solution: ${}^{9}C_{5} = 126$ different teams can be selected.

24. (a) Solution: There are ${}^{20}P_{5} = 1,860,480$ ways to choose a chairperson, assistant chairperson, treasurer, community advisor, and record keeper.
(b) Solution: There are ${}^{16}P_{4} = 43,680$ ways coach can select students to compete in the race.
(c) Solution: There are ${}^{15}P_{2} = 210$ ways student can be chosen for these 2 positions.

25. (a) Solution: ${}^{5}C_{1} * {}^{3}C_{2} * {}^{4}C_{1} * {}^{6}C_{3} = 1200$ Sandwiches can be made of 1 meat, 2 bread, 1 cheese, and 3 condiments.
(b) Solution: There are ${}^{12}C_{5} = 792$ different toppings on ice cream.
(c)
(i) Solution: With repetition: 1*10*10*10*10 = 10,000 Zip codes Without repetition: 1*9*8*7*6 = 3024 Zip codes
(ii) Solution: With repetition: 2*10*10*10*10 = 20,000 Zip codes Without repetition: 2*9*8*7*6 = 6048 Zip codes

26. (a) Solution:
A = Strings begins with three 0s = $2^7 = 128$ B = Strings end with two 0s = $2^8 = 256$ $A \cap B = 2^5 = 32$
$AUB = A + B - A \cap B = 128 + 256 - 32 = 352$.
(b) Solution:
A = Strings begins with 0s = $2^4 = 16$ B = Strings end with two 1s = $2^3 = 8$ $A \cap B = 2^2 = 4$
$AUB = A + B - A \cap B = 16 + 8 - 4 = 20$.
(c)
(i) Solution: With repetition: 26*26*10*10*10*10 = 6,760,000 Without repetition: 26*25*10*9*8*7 = 3,276,000
(ii) Solution: With repetition: 5*5*5*5*5 = 3125 Without repetition: 5*4*3*5*4 = 1200

27. (a) Solution: The first letter of each last name are the pigeonholes, and the letters of the alphabet are pigeons. By the generalized pigeonhole principle, $\lceil 30/26 \rceil = 2$. So, there are at least two students, have last names that begin with the same letter.
(b) Solution: By the generalized pigeonhole principle, $\lceil 8008278/1000000 \rceil = 9$.
(c) Solution: The 38 time periods are the pigeonholes, and the 677 classes are the pigeons. By the generalized pigeonhole principle there is at least one time period in which at least $\lceil 677/38 \rceil = 18$ classes are meeting. Since each class must meet in a different room, we need 18 rooms.

28. (a) (i) Solution: From binomial theorem, it follows that coefficient is: ${}^{n}C_{r} = {}^{11}C_{5} = 462$.
(ii) Solution: From binomial theorem, it follows that coefficient is: ${}^{n}C_{r} = {}^{24}C_{17} (2)^7 (-1)^{17} = - 44,301,312$.

(b) (i) Solution:

1 ← row 0
1 1
1 2 1
1 3 3 1
1 4 6 4 1 ← row 4
1 5 10 10 5 1 ← row 5
1 6 15 20 15 6 1 ← row 6

$\binom{0}{0}$
$\binom{1}{0}\ \binom{1}{1}$
$\binom{2}{0}\ \binom{2}{1}\ \binom{2}{2}$
$\binom{3}{0}\ \binom{3}{1}\ \binom{3}{2}\ \binom{3}{3}$
$\binom{4}{0}\ \binom{4}{1}\ \binom{4}{2}\ \binom{4}{3}\ \binom{4}{4}$
$\binom{5}{0}\ \binom{5}{1}\ \binom{5}{2}\ \binom{5}{3}\ \binom{5}{4}\ \binom{5}{5}$
$\binom{6}{0}\ \binom{6}{1}\ \binom{6}{2}\ \binom{6}{3}\ \binom{6}{4}\ \binom{6}{5}\ \binom{6}{6}$

(ii) Solution:

$$= 1(3x)^4(-2)^0 + 4(3x)^3(-2)^1 + 6(3x)^2(-2)^2 + 4(3x)^1(-2)^3 + 1(3x)^0(-2)^4$$

$$= 3^4 x^4 + 4\cdot 3^3\cdot(-2)\,x^3 + 6\cdot 3^2\cdot 4\,x^2 + 4\cdot 3\cdot(-8)\cdot x + 16$$

$$\boxed{= 81x^4 - 216x^3 + 216x^2 - 96x + 16}$$

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

(c) (i) Solution:

$= {}^6C_0(2x)^{6-0}(y)^0 + {}^6C_1(2x)^{6-1}(y)^1 + {}^6C_2(2x)^{6-2}(y)^2 + {}^6C_3(2x)^{6-3}(y)^3 + {}^6C_4(2x)^{6-4}(y)^4 + {}^6C_5(2x)^{6-5}(y)^5 + {}^6C_6(2x)^{6-6}(y)^6$

$(2x+y)^6 = 64x^6 + 192\,x^5y + 240\,x^4y^2 + 160\,x^3y^3 + 60\,x^2y^4 + 12\,xy^5 + y^6$

(ii) Solution:

$= {}^7C_0(3a)^{7-0}(-2b)^0 + {}^7C_1(3a)^{7-1}(-2b)^1 + {}^7C_2(3a)^{7-2}(-2b)^2 + {}^7C_3(3a)^{7-3}(-2b)^3 + {}^7C_4(3a)^{7-4}(-2b)^4 + {}^7C_5(3a)^{7-5}(-2b)^5 + {}^7C_6(3a)^{7-6}(-2b)^6 + {}^7C_7(3a)^{7-7}(-2b)^7$

$= (3a-2b)^7 = 2187a^7 - 10206\,a^6b + 20412\,a^5b^2 - 22680\,a^4b^3 + 15120\,a^3b^4 - 6048\,a^2b^5 + 1344\,ab^6 - 128\,b^7$

29. (a) (i) Solution: There are 36 students. They can be put in a row in 36! ways.

(ii) Solution: You need to have an ordered arrangement of 7 out of 36 students. The number of such arrangements is P (36,7).

(iii) Solution: You need to have an ordered arrangement of all 20 women and ordered arrangement of all 16 men. By the product rule, this can be done in 20! * 16! ways.

(b) Solution: No pitcher * Last 5 pitcher in = ${}^8P_4 * {}^5P_5 = 201{,}600$.

(c) Solution: ${}^{12}C_0 + {}^{12}C_1 + {}^{12}C_2 + {}^{12}C_3 = 299$.

30. (a) Solution: Expression = $({}^{15}C_3)\ ({}^{12}C_3)\ ({}^7C_3)\ ({}^5C_3)\ ({}^9C_3)\ ({}^{10}C_3) = 3.531 \times 10^{11}$.

(b) Solution:

Let N be 97 students and K be 10 different grades.

$\lceil N/K \rceil = \lceil 97/10 \rceil = 10$. So, students 10 will earn the exact same grade.

(c) Solution: Squad is consisting of 15 players so, ${}^{15}P_1 \times {}^{14}P_1 \times {}^2P_1 = 420$.